

|                                                       |                                      |
|-------------------------------------------------------|--------------------------------------|
| Program : <b>Diploma in Wood and Paper Technology</b> |                                      |
| Course Code : <b>4121</b>                             | Course Title: <b>Wood Products I</b> |
| Semester : <b>4</b>                                   | Credits: <b>4</b>                    |
| Course Category: <b>Program Core</b>                  |                                      |
| Periods per week: <b>4 (L:3, T:1, P:0)</b>            | Periods per semester: <b>60</b>      |

### Course Objectives:

- To familiarize students with veneer manufacturing process in plywood industry, possibility of solid and laminated wood, and wood bending processes

### Course Prerequisites:

| Topic              | Course code | Course name                      | Semester |
|--------------------|-------------|----------------------------------|----------|
| Structure of wood  |             | Botany and Anatomical Structures | 2        |
| Processing of wood |             | Wood processing                  | 3        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                   | Duration (Hours) | Cognitive Level |
|-----------------|---------------------------------------------------------------|------------------|-----------------|
| CO1             | Outline the raw materials used for plywood production         | 15               | Understanding   |
| CO2             | Summaries veneer processing and manufacture of plywood        | 14               | Understanding   |
| CO3             | Interpret grading of plywood , testing and properties         | 15               | Applying        |
| CO4             | Compare solid wood bending and laminated wood bending process | 14               | Applying        |
|                 | Series Test                                                   | 2                |                 |

**CO - PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     |     |     |     |     |
| CO2             | 3   |     |     |     |     |     |     |
| CO3             |     | 3   |     |     |     |     |     |
| CO4             | 3   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| Module Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Description                                                   | Duration (Hours) | Cognitive Level |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|------------------|-----------------|
| CO1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Outline the raw materials used for plywood production</b>  |                  |                 |
| M1.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Recall History and development of Plywood manufacturing       | 1                | Remembering     |
| M1.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Explain Processing of Logs Prior to Veneering                 | 5                | Understanding   |
| M1.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Illustrate rotary cutting and Splicing of veneer              | 4                | Applying        |
| M1.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Outline Drying and Splicing of Veneer                         | 5                | Applying        |
| <b>Contents:</b><br>Plywood, veneer, flitches, peeling, slicing and drying - advantage of plywood over solid wood - history of plywood manufacture - factors considering in the selection of logs and its storage and transport - end coating and end protection devices - softening treatment of logs prior to boiling and slicing - flitches and their preparation - tangential and radial cutting - veneer slicing and methods - vertical slicing machine - rotary peeling - nose bar, knife arrangement on peeling lathe - Illustrate schematic diagram of log with respect to knife and nose bar - handling of veneers and clipping - advantage of veneer drying - Jet dryer, roller dryer, band dryer, breather and Kiln dryer - processing of veneers - storage and handling of veneers from peeling - joining and splicing machine- cross feed splicer |                                                               |                  |                 |
| CO2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Summarise veneer processing and manufacture of plywood</b> |                  |                 |
| M2.01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Illustrate repairing of Veneer                                | 3                | Understanding   |
| M2.02                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Illustrate Veneer matching                                    | 3                | Applying        |
| M2.03                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Explain gluing of veneer                                      | 4                | Understanding   |
| M2.04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Outline assembling of veneer                                  | 4                | Understanding   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Series Test - I                                               | 1                |                 |

**Contents:**

Repair of veneers - picus joint, scarf joint, Ibus joint - veneer matching - four piece matching, book matching side to side matching, end to end matching cathedral matching - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - adhesive, adhesion, adherent - glues - glue mixing - roller spreading with diagram, brushing, spraying - pre pressing and its advantages - cold press - hot press - gauge pressure - conditioning - hot press gluing faults such as blister, bleed through, warp age, starved joint, spotty bond, weak bond face checking, overlaps and core gaps - dimensioning and sanding of plywood - trimming - standard plywood specifications - drum sander, belt sander, wide belt sander - abrasive materials

|            |                                                             |   |               |
|------------|-------------------------------------------------------------|---|---------------|
| <b>CO3</b> | <b>Interpret grading of plywood, testing and properties</b> |   |               |
| M3.01      | Illustrate Pressing and Pre-Pressing                        | 5 | Applying      |
| M3.02      | Explain Dimensioning of Plywood                             | 5 | Understanding |
| M3.03      | Outline Grading of Plywood                                  | 5 | Understanding |

**Contents:****Grading of Plywoods**

Physical properties - density, moisture - content, hygroscopic properties, absorption, swelling and shrinkage Mechanical properties - elasticity, rigidity tensile strength, bending Thermal conductivity and acoustical properties Testing of plywood as per BIS code - determination of density, moisture content - glue shear strength, knife test, tensile strength, compressive strength, impact strength bending strength, nail and screw holding power, determination of fire resistance, mycological test, acidity and alkalinity Speciality plywoods, Tea chest plywood, marine plywood, shuttering plywood, fire proof plywood, air craft plywood, preservative treated plywood, metal faced plywood, moulded Plywood.

|            |                                                                      |   |             |
|------------|----------------------------------------------------------------------|---|-------------|
| <b>CO4</b> | <b>Compare solid wood bending and laminated wood bending process</b> |   |             |
| M4.01      | Recall Wood Bending                                                  | 3 | Remembering |
| M4.02      | Explain Solid Wood Bending                                           | 6 | Applying    |
| M4.03      | Outline laminated wood bending                                       | 5 | Applying    |
|            | Series Test - II                                                     | 1 |             |

**Contents:**

Bending - General Consideration in bending - steaming - cold bending and Hot Bending - simple 'U' bend with figure - Re entrant and 'S' type bend with figure - end stops - chair leg bending - Machine Bending - Machine bending - Rope and windless Machine with figure - Lever arm Machine with figure - Revolving Table Machine - setting technique - Movement of Bends - laminated bending - selection and Preparation of laminates - Pre bending - laminated wood bending by male & female forms - split female forms, by inflated flexible hose - two plane bends

Curved laminated Members- Movement and distortion of laminated bends- Plywood bending

**Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                                                             |
|------------|--------------------------------------------------------------------------------------|
| T1         | F.P. Kollmann, Principles of wood science and technology Vol.I, II - Springer Verlag |
| R1         | W.C. Stevens ,Solid and laminated wood bending - Majesty's stationary office         |
| R2         | Lecturer note on plywood technology Vol.I, II - IPIRI, Bangalore.                    |

**Online Resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://youtu.be/o7YnSV8P-IQ">https://youtu.be/o7YnSV8P-IQ</a>                               |
| 2            | <a href="https://www.youtube.com/watch?v=MzDaQ8z3jII">https://www.youtube.com/watch?v=MzDaQ8z3jII</a> |
| 3            | <a href="https://www.youtube.com/watch?v=cuMJ6c5r7UQ">https://www.youtube.com/watch?v=cuMJ6c5r7UQ</a> |